How are mass extinctions shaping biodiversity?

Confirmation Review

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Part I

Abstract

Part II

Background Review

1 Morphology, Disparity and Extinction

1.1 Mass extinctions

Extinction is a fundamental evolutionary process that, alongside speciation, continuously shapes patterns of biodiversity through deep time; over 99% of all species that have ever lived are now extinct (Raup, 1981). However, extinction rates are not constant and occur in pulses across background taxonomic turnover (Raup and Sepkoski, 1982). *Mass extinctions* are specific events that wipe out over 75% of species and have happened five times over the last 500 million years (Raup and Sepkoski, 1982). On a macroevolutionary scale, mass extinctions mark the most dramatic species composition transitions in Earth history (Jablonski,

2001). Entire branches are trimmed from the tree of life, while the evolutionary trajectories of survivors are altered in the wake of a transformed global biota (Hull, 2015).

1.2 Rebound dynamics

The aftermath of this drastic species turnover is typically described as 'recovery', yet Jablonski and Edie (2025) describe this as a misnomer, since this implies a return to the prior pre-extinction state. Instead, biodiversity rebounds are multifaceted and the response from the survivors of a mass extinction event may take several forms, transforming the post-extinction biota (Hull, 2015). For example, trends observed in different clades pre-mass extinction may continue unperturbed, including escalations (Hansen et al., 1999) and declines (Raup, 1981). Alternatively, the aftermath of a mass extinction can trigger diversification as surviving clades are offered ecological opportunity, filling the vacant niches left by extinct lineages.

Rebound dynamics can differ by the metric of diversity considered. A discordance between taxonomic and morphological diversity, which can be considered a proxy of ecological diversity, is common in empirical macroevolutionary studies. Intuitively, a diversifying clade would exhibit simultaneous taxonomic and phenotypic diversification, which are expected to follow a pattern of early high rates that decrease through time as niches become saturated (Schluter, 2000). However, empirical evidence often finds that peak taxonomic and morphological diversity are reached at different points in a clade's history (Foote, 1993; Hopkins, 2013; Ruta et al., 2013; Wan et al., 2021). This discordance has been observed across mass extinction events, such as in graptolites across the end-Ordovician mass extinction (Bapst et al., 2012), or therapsids across the Permian-Triassic mass extinction (Ruta et al., 2013). It has been suggested that functional diversity may be density-independent across mass extinction events (Edie et al., 2018). Therefore, understanding the rebound dynamics of clades based solely on taxonomic diversity metrics may not encapsulate the full complexity of rebound dynamics (Puttick et al., 2020).

1.3 Disparity

This phenotypic diversity, often called *disparity*, has been widely used to investigate patterns of evolution in extant and extinct clades (Wills et al., 1994). Disparity refers to the extent of morphological diversity among taxa, typically quantified as the distribution or spread of those taxa within a morphospace — a multidimensional mathematical space configured by morphological similarity (where similar looking species are positioned closer together) (Guillerme et al., 2020). Changes in morphospace through time are often closely tied to shifts in ecological diversity (Lowery and Fraass, 2019; Budd, 2021). For example, expansion in morphospace, measured by variance-based metrics, may indicate a clade expanding its niche breadth during an adaptive radiation (Foote, 1994; Raup, 1994; Sidlauskas, 2008). Alternatively, a clade may move position within the morphospace, reflecting shifts in ecological niche (Hopkins, 2016).

1.4 Extinction selectivity

Mass extinctions pose numerous scenarios of disparity (Figure 1). In adaptive scenarios, disparity may increase in clades that diversify during the period of ecological opportunity following a mass extinction. Random extinction will reduce the total number of taxa, but have little effect on sized-based disparity metrics. Selective extinction, whereby taxa are more susceptible to extinction due to particular trait combinations, will have a detectable effect on variance-based measures (Foote, 1997; Korn et al., 2013). Korn et al. (2013) proposed several patterns to describe different selection processes: lateral (asymmetric selection that trims one side of the morphospace), marginal (symmetric selection that removes peripheral taxa) and non-selective (random process independent of the variables defining the morphospace). Work on contemporary extinctions has also investigated extinction selectivity, with certain trait combinations making extant taxa more susceptible to extinction (Ali et al., 2023; Matthews et al., 2024).

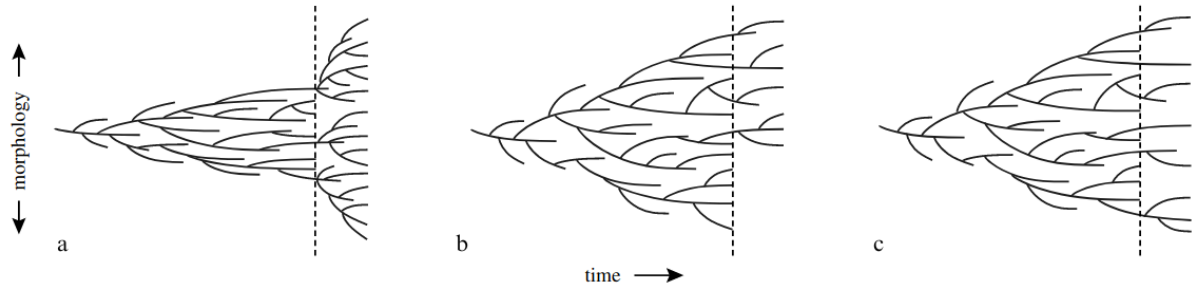


Figure 1: Three scenarios that may occur within a clade across a mass extinction (denoted by dashed line). a) adaptive radiation. b) selective extinction. c) non-selective extinction. Adapted from Grossnickle and Newham (2016).

2 Eutherian disparity across K-Pg Mass Extinction

2.1 Suppression hypothesis

The rebound dynamics of eutherian mammals (placentals and extinct relatives) across the K-Pg mass extinction is one of the most well-studied yet contentious inquiries of macroevolutionary biology. Eutherian mammals prior to the K-Pg boundary have been traditionally considered depauperate, small sized and limited in ecological variation, which contrasts the diversity seen in extant placentals. The widely held view until recently, known as *the suppression hypothesis*, follows that the extinction of non-avian dinosaurs released eutherian mammals from constraints of predation and competition, stimulating a radiation at the onset of the Cenozoic (Osborn, 1902; Alroy, 1999; Codron et al., 2013; O'Leary et al., 2013). The most significant morphological evidence for the suppression hypothesis in the fossil record concerns a shift in body size disparity (Alroy, 1999). Mesozoic mammals generally weighed less than 100 g and rarely over 5 kg (Benevento

et al., 2023). The maximum body mass of mammals increased exponentially and by roughly three orders of magnitude by the late Eocene, which is consistent across continents and taxonomic groups (Cooper and Purvis, 2010; Smith et al., 2010; Raia et al., 2013; Slater, 2013).

2.2 Early rise hypothesis

While the suppression hypothesis is strongly supported by studies that have investigated changes in body size (Alroy, 1999; Slater, 2013), body size is limited as a biodiversity metric for ecological variation since it can be influenced by several factors that may reflect stochastic rather than adaptive changes (Grossnickle, 2020). Recent studies instead focusing on morpho-functional traits (intrinsically linked to ecological function), such as jaw and dentition morphology, have instead suggested that eutherian diversity was attained through successive waves of diversification, starting far earlier than the K-Pg mass extinction (Grossnickle et al., 2019). The early rise hypothesis suggests an ecological diversification that occurred concurrently with the diversification of angiosperms and insects during and after the Cretaceous Terrestrial Revolution (125–80 Ma) (Figure 2) (Meredith et al., 2011; Benton et al., 2022). Ecological variation within Mesozoic mammals is not limited to diet; diversity in locomotor styles was reached as far back as the Jurassic in early mammaliaform groups, including early eutherians (Chen et al., 2019).

2.3 Delayed rise hypothesis

Other studies have pointed towards the Paleocene-Eocene Thermal Maximum (PETM), an episode of elevated global temperatures 56 Ma that widened the latitudinal range of angiosperm-dominated forest, which in turn spurred the dispersal and radiation of eutherians (Figure 2) (Bininda-Emonds et al., 2007). Benevento et al. (2019) found that eutherian mammals had the most significant expansion in jaw morphospace at the onset of the Eocene. While these alternative hypotheses do not necessarily refute the influence of the K-Pg mass extinction, they instead suggest the extinction accentuated evolutionary trajectories already underway.

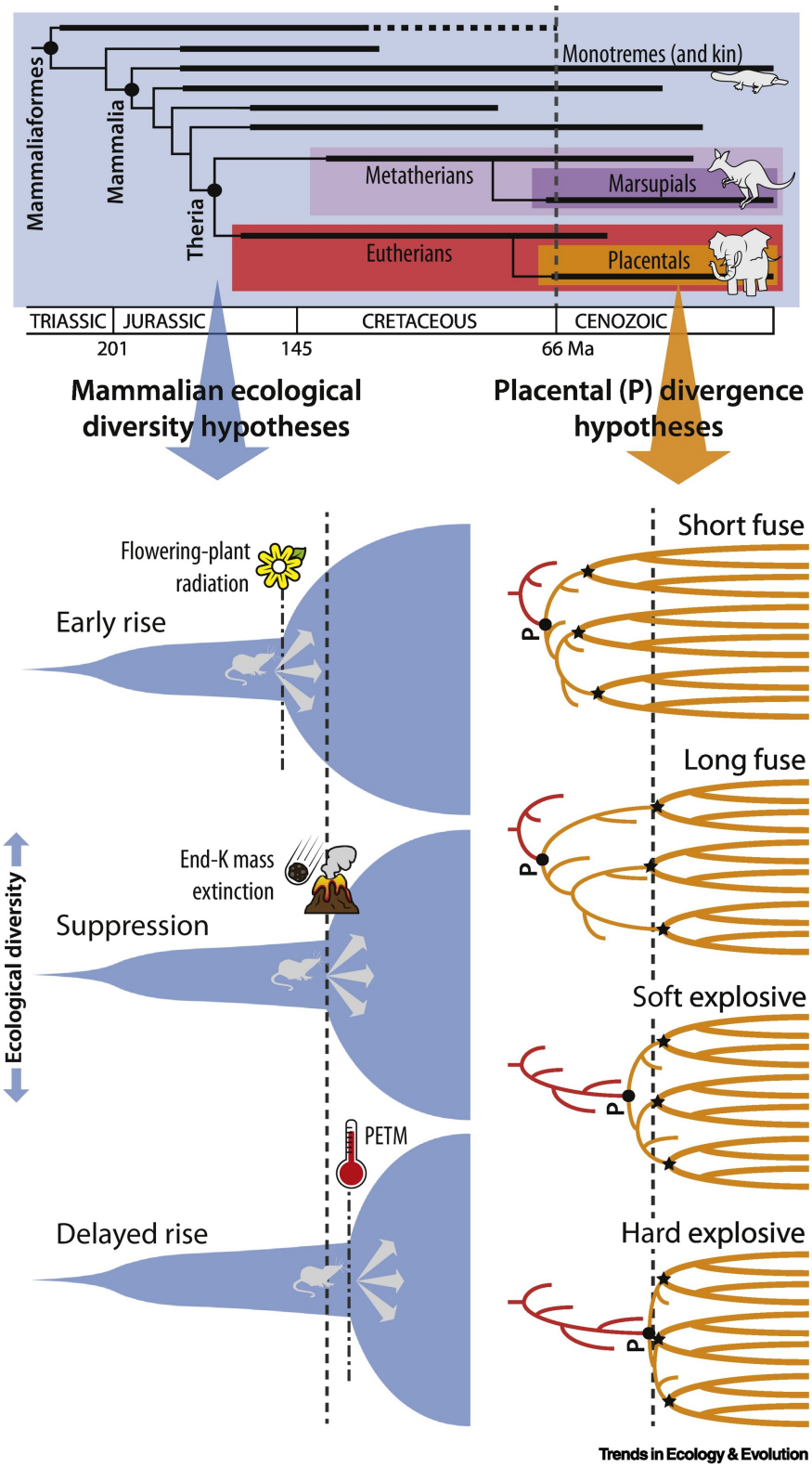


Figure 2: The contrasting ecological diversification triggers and placental origin hypotheses. Adapted from Grossnickle et al. (2019).

2.4 Phylogenetic hypotheses

The origin and tempo of lineage diversification in placental mammals has been a point of contention for decades, meaning there is no undisputed phylogenetic topology for eutherian mammals. Therefore, several studies investigating eutherian disparity have opted to time-bin the disparity data points by stratigraphy when assessing disparity through time (Grossnickle and Polly, 2013; Wilson, 2013; Grossnickle and Newham, 2016; Benevento et al., 2019). While this approach may be necessary in the absence of a well-resolved phylogeny, it inherently assumes a punctuated equilibrium model of morphological evolution. This assumption can be particularly problematic when assessing disparity across mass extinction events (Guillerme and Cooper, 2018). The instantaneous nature of the K-Pg extinction requires a framework capable of capturing morphological changes at fine temporal scales. Stratigraphic time binning, which aggregates data over broad intervals, risks obscuring critical, transient shifts in disparity that may occur within shorter durations, e.g. 1 Myr following the extinction. Incorporating a phylogeny means a time-slicing method can be used (Guillerme and Cooper, 2018), which removes the punctuated assumption and can reveal more temporally refined patterns. There are currently four phylogenetic hypotheses for placental mammals (Figure 2):

2.4.1 Hard explosive model

To date, there are no unequivocal crown placental fossils prior to the K-Pg boundary (Alroy, 1999; Davies et al., 2017). This palaeontological evidence has given rise to the hard explosive model, which posits that the origin and diversification of placental lineages took place at/after the K-Pg boundary (Figure 2). The stark contrast in taxonomic richness (11 Late Cretaceous eutherians compared to 139 in the Paleocene) has been shown to not be an artefact of poor Cretaceous sampling (Davies et al., 2017). Therefore, until a definitive Late Cretaceous placental mammal has been found, the hard explosive model cannot be ruled out.

2.4.2 Long fuse model

In contrast, divergence estimates using molecular data suggest a Late Cretaceous origin for placental mammals. Two further models have been developed from molecular analyses, namely the long fuse and short fuse models (Figure 2). The long fuse model implies a placental origin coinciding with the Cretaceous Terrestrial Revolution approximately 120-100 Ma (Foley et al., 2016). This model also posits that most interordinal cladogenesis occurred prior to the K-Pg, followed by intraordinal diversification during the Paleocene (Archibald and Deutschman, 2001). This model has garnered support from the majority of molecular-based estimates, notably recent genomic studies (Meredith et al., 2011; dos Reis et al., 2012; Foley et al., 2016; Carlisle et al., 2023).

2.4.3 Short fuse model

The short fuse model has been recovered relatively little compared to the long fuse model. Like the long fuse model, the model finds a Late Cretaceous origin of Placentalia, but differs with intraordinal diversification

taking place prior to K-Pg. The most notable study to have recovered the short fuse model is that of Bininda-Emonds et al. (2007), which also found the K-Pg extinction had little impact on placental diversification rates. However, this study employed local molecular clocks and a pure birth model for divergence times; more recent relaxed clock studies have refuted this finding and supported the long-fuse model.

2.4.4 Soft explosive model

Recent attempts to close the gap between fossil and molecular estimates have culminated in the soft explosive model. Phillips (2016) argues that divergence times are overestimated when calibrating nodes for large and/or long-lived taxa with slow molecular rates, which contrasts the small early placental taxa that likely had faster molecular rates. By removing these calibrations, Phillips (2016) recovers superordinal cladogenesis in the Late Cretaceous, but the remainder of diversification near or after the K-Pg boundary (Figure 2).

3 Disparity Analysis Methods

The conflicting findings of eutherian disparity across the K-Pg boundary raise the question: why do disparity analyses aiming to answer the same question arrive at different conclusions? Addressing this requires a critical look at the methodological underpinnings of disparity analyses.

For the purposes of this review, I focus on three key methodological choices: (1) trait data, (2) disparity metric, (3) ancestral state estimation.

3.1 Types of trait data

The morphological data used in disparity analyses may fall into one of the following categories:

- **Discrete characters**, i.e. coding presence/absence or characteristic of traits. Discrete character matrices allow the inclusion of morphological features that are only present in a subset of the taxa. These datasets are often originally collected for phylogenetics and repurposed for disparity.
- **Continuous characters**, i.e. lengths of limbs, body mass, etc. More suited to investigations on a specific trait, rather than whole organism disparity.
- **Geometric morphometric**, i.e. homologous landmark data (e.g. Procrustes coordinates). More common for analyses at lower taxonomic levels, since need for homologous features in all specimens.

Given the varying scope of morphological variation that these contrasting approaches characterise, it is expected that they capture different aspects of disparity. For example, molar geometric morphometric data will mainly reveal variation related to diet/feeding ecology, whereas discrete character matrices of cranial and postcranial features may reveal variation in whole-organism morphology. Correlation between the outputs of these two approaches cannot be assumed, highlighting the discrepancies between disparity

analyses. For example, Halliday and Goswami (2016), using a cladistic data matrix of dental, cranial and postcranial characters, found contrasting patterns of disparity across K-Pg compared to studies that used geometric morphometric analysis of molars (Wilson, 2013; Grossnickle and Newham, 2016). Furthermore, the overwhelming phylogenetic signal in cladistic discrete datasets, i.e. Halliday and Goswami (2016), may bury important functional variation contained within (Anderson and Friedman, 2012). Characters used during phylogenetics are selected to establish branching orders and differentiate taxa into distinct clades, which may artificially inflate disparity at nodes of cladogenesis (Gerber, 2019).

3.2 Disparity metrics

The contrasting findings may also be explained by different disparity metrics used between studies. For example, Halliday and Goswami (2016) found contrasting patterns of eutherian disparity between outputs using the sum of ranges compared to mean pairwise distance and sum of variances. Puttick et al. (2020) highlighted the importance of employing several different disparity metrics to explore different ways in which the morphospace changes across a mass extinction. However, several studies investigating eutherian disparity have opted to use only a single disparity metric, often variance-based metrics such as sum of variances/ranges (Grossnickle and Polly, 2013; Wilson, 2013; Grossnickle and Newham, 2016). Puttick et al. (2020) highlighted the importance of employing several disparity metrics to explore different ways in which the morphospace changes across a mass extinction.

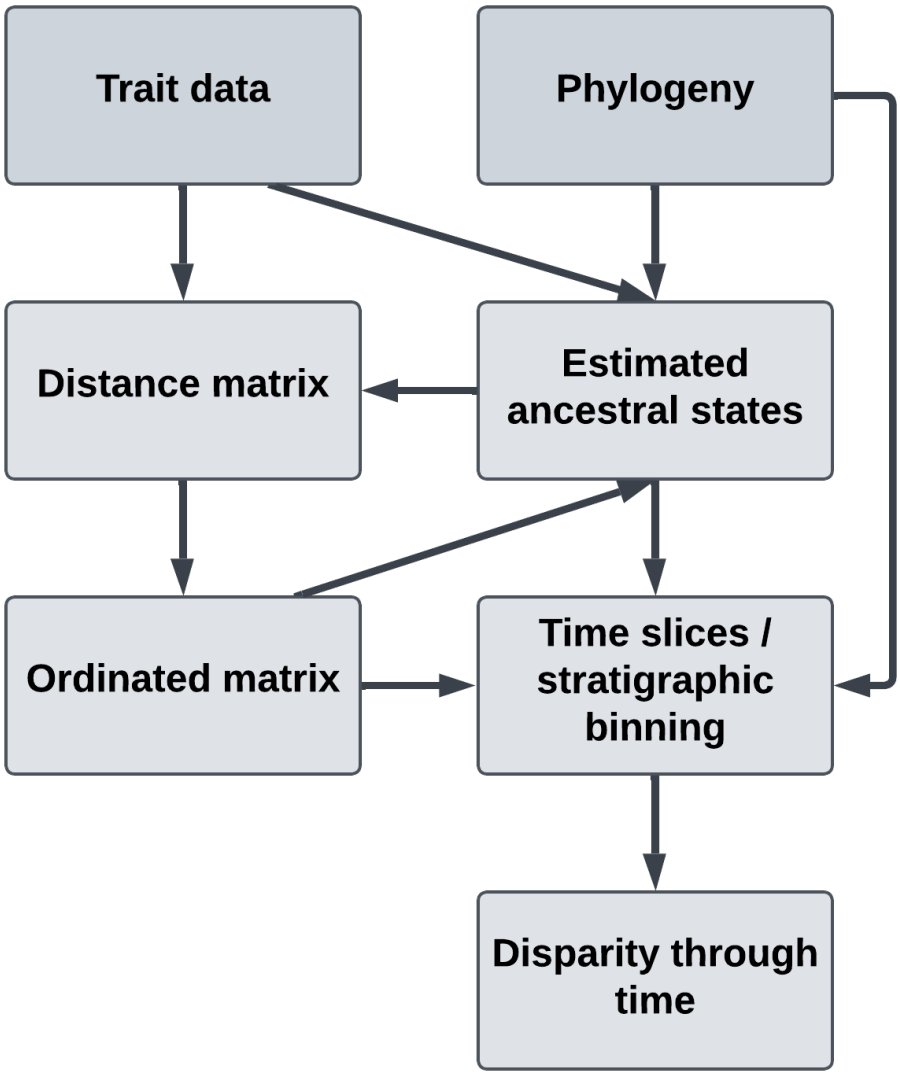


Figure 3: Disparity pipeline (N.B. this is specifically for discrete characters. Continuous characters would bypass the distance matrix after ASE or without ASE). As shown there are several routes to disparity, including bypassing ancestral state estimation (ASE), pre-ordination ASE, post-ordination ASE

4 Ancestral state estimation

The poor resolution of the fossil record is an inevitable limitation affecting disparity analyses, since only a small biased portion of a clade's evolutionary history is preserved. The incorporation of a phylogeny together with trait data enables a switch from simply observing a pattern (points in morphospace) to inferring a process through time, through the use of ancestral state estimation (ASE) (Lloyd, 2018).

4.1 Application of ASE

Ancestral state estimation (ASE) is the broad method of estimating trait value(s) in a set of common ancestors given a set of observed data for tips on a phylogenetic tree (Schluter et al., 1997; Cunningham et al., 1998; Yang, 2006; Revell, 2025). Trait data may be continuous, such as body size, or discrete, such as diet type or presence/absence of a character. The application of ASE has been widely utilised in phylogenetic comparative methods, including disparity.

Intuitively, ASE is appealing for those studying deep-time patterns of disparity. Incorporating a phylogeny into disparity analyses will unavoidably introduce unsampled 'ghost lineages' (Brusatte et al., 2011). A ghost lineage is the hypothetical ancestor that is not observed in the fossil record, but can be inferred to have existed due to the presence of other fossils/extant taxa. This can be best explained as the branch on a phylogeny between two observed taxonomic units (Figure 4).

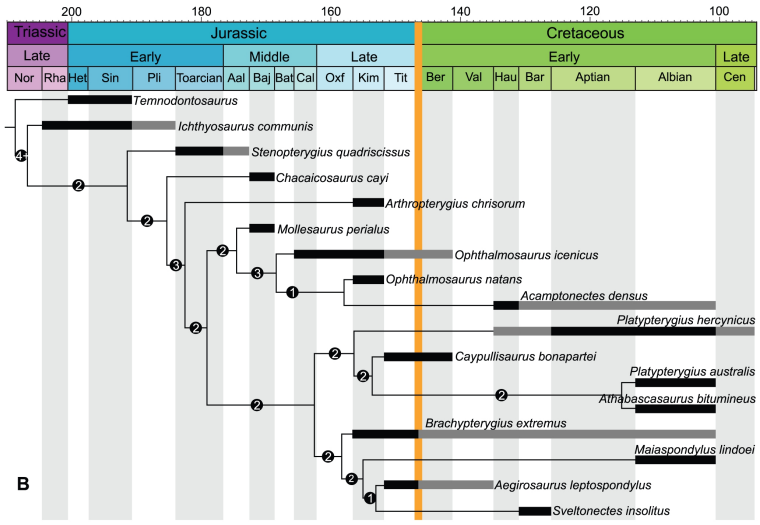


Figure 4: Phylogeny of ichthyosaurs showing observed stratigraphic range for each species (thick lines) and inferred ghost lineages (thin lines) (Fischer et al., 2012).

ASE has been employed as a way of inferring the traits of these hypothetical species based on the traits of their descendents. During ASE, the estimated traits for nodes of a phylogeny are used as a way of sampling these ghost lineages. This fills in the gaps in the fossil record and alleviates the bias associated with uneven sampling across the phylogenetic tree (Friedman, 2009; Brusatte et al., 2011).

4.1.1 Likelihood methods

Likelihood methods for ancestral state estimation are based on a unifying statistical framework that treats the internal node states as parameters to be estimated, seeking to maximize the probability of the observed tip states given a topology and a model of evolution (Joy et al., 2016). The central tenet of the approach is the likelihood function, which integrates information from the topology, which reflects the evolutionary

history among taxa, and the model of trait evolution (which specifies how traits evolve along the branches). By systematically exploring the parameter space through numerical optimization, the method search for the ancestral states that maximize the likelihood function. Importantly, the same general likelihood framework underpins different types of ancestral state estimation, regardless of whether the traits are continuous or discrete. The core difference between continuous and discrete character estimation is the evolutionary model that is specified.

4.1.2 Continuous characters

Continuous trait evolution is typically modelled under a Brownian motion (BM) model of evolution (Felsenstein, 1985; O'Meara et al., 2006). Brownian motion models trait evolution as a directionless, stochastic process under which continuous characters undergo changes along the branches of a phylogeny. These evolutionary changes come from a Gaussian distribution with a mean of 0 and variance proportional to the product of the rate of evolution and time elapsed ($\sigma^2 t$). Therefore, the expected amount of trait divergence between any two taxa is proportional to the amount of shared evolutionary history according to the phylogeny.

Under BM, the vector ($\mathbf{x}$) of observed tip trait values follows a multivariate normal distribution (MVN) and has an expected value that is equal to the root state x_0.

$$\boldsymbol{x} \sim \mathrm{MVN}(x_0,\, \sigma^2 \boldsymbol{C})$$

$\boldsymbol{C}$ is a variance-covariance matrix that captures the shared ancestry of the taxa, defined by the tree height from the root to each tip (diagonal elements $\boldsymbol{C}_{ii}$) and the tree height above the root of the most recent common ancestor shared by each pair of tips (off-diagonal elements $\boldsymbol{C}_{ij}$).

When estimating ancestral states, the $\boldsymbol{C}$ matrix is expanded to include the shared history of the internal nodes. The evolutionary rate, σ^2, and root state x_0, are unknown parameters that can be estimated using Maximum Likelihood to give the multivariate normal distribution. The trait values at internal nodes are then calculated as the conditional expectations within this distribution. This means each node's estimated trait value is effectively a weighted average of the tip values, with the weights determined by the amount of shared evolutionary history (as captured by $\boldsymbol{C}$).

4.1.3 Discrete characters

Ancestral state estimation of discrete characters are modelled using a continuous-time Markov chain process, known as the Mk model (Pagel, 1994; Yang, 2006). This process is specified by a $k \times k$ matrix, $\mathbf{Q}$, which gives the instantaneous transition rates (q) between states for each character.

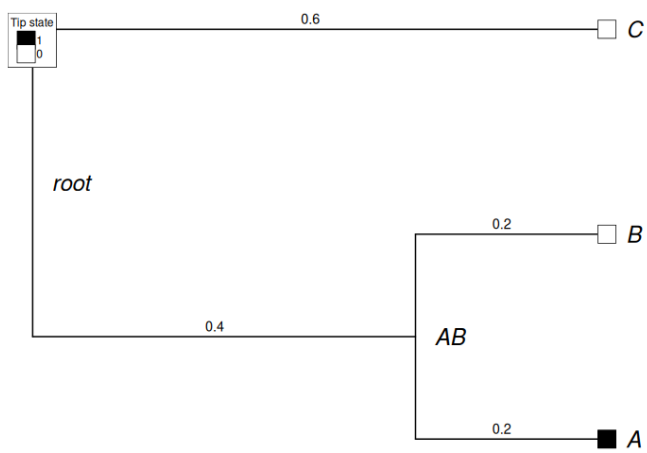
$$\mathbf{Q} = \begin{bmatrix} -q_{0,1} & q_{0,1} \\ q_{1,0} & -q_{1,0} \end{bmatrix}$$

Markov chain transition matrices has the necessary properties that all off-diagonal entries are non-negative, all diagonal properties are non-positive and the sum of each row is zero. This is because in a continuous-time Markov process, probability flows out of a state (diagonal) and into other states (off-diagonal) and so the diagonal term represents the total rate of leaving a state.

It is this $\mathbf{Q}$ matrix that is estimated under Maximum Likelihood. Given a phylogeny, the probability of each transition at any given interval of time (t) can be calculated by

$$\mathbf{P} = \exp(\mathbf{Q}t)$$

This gives the transition probability matrix, where each entry is the probability of transitioning from one state to another over time t. The most popular method is *marginal* estimation, which involves proceeding from node to node on the phylogeny, and, at each node, the probability of being in each state is estimated, given the observed data. These likelihoods are rescaled by their sum so that they add up to 1.0, and so represent an empirical Bayes posterior probability (Yang, 2006; Revell, 2025). Since this can become extremely complicated as the phylogeny increases in size, in practice functions use algorithms such as Felsenstein's pruning algorithm (Felsenstein, 1981).



So if we want to calculate probability of each transition along the tree branch from root to node AB, $t = 0.4$.

One further probability to consider is the 'root prior', which is essentially whether the root node of the tree was in state 0 or 1. This is typically set with a flat root prior $\pi_0 = \pi_1 = 0.5$.

These marginal likelihoods are then scaled by their sum, giving $P(0,\ 1) = [0.68,\ 0.32]$. In marginal estimation, this is calculated node by node. Since this can become extremely complicated as the phylogeny increases in size, in practice functions use algorithms such as Felsenstein's pruning algorithm (Felsenstein,

1981).

4.2 Challenges of ASE

'Character state reconstructions can provide a powerful mechanism for studying many facets of the evolutionary process. However, the zeal with which these techniques are sometimes advocated belies the complexity of the problem' (Swofford and Maddison, 1992).

4.2.1 Uncertainty in estimates

A critical issue within ancestral state estimation is that estimates tend to have broad uncertainty (Schluter et al., 1997). This uncertainty is not uniform across the tree: it typically increases towards root of the tree, where direct information from observed tip data becomes increasingly diffuse (Gascuel and Steel, 2014). For discrete characters, this uncertainty manifests as inconclusive scaled likelihoods, i.e., state 0 = 0.51, state 1 = 0.49. For continuous characters, uncertainty is wide confidence intervals.

How this uncertainty in estimations should be dealt with is equivocal. Despite the encouragement for uncertainty to be carried forward in comparative analyses (Silvestro et al., 2015), most computational implementations of ASE return point estimates. For continuous characters, this will be the median value of the confidence interval and for discrete characters, it will be the most probable state according to a threshold (see later). By ignoring uncertainty, point estimates can lead to overconfidence in results and erroneous inferences in downstream analyses (such as down the disparity pipeline), all while obscuring potentially 'accurate' uncertainty.

For discrete characters, there is a level of subjectivity in how to collapse the states into the most probable based on their scaled likelihood. It is common to have a threshold implementation in ASE software, whereby the user can set a scaled likelihood at which point states are collapsed to a point estimate, or left as uncertain, which is denoted as '0/1', for example. However, these thresholds are often arbitrarily set with no guidance of what is the 'correct' value (Smith and Donoghue, 2022; Casali et al., 2023). Much of the subjectivity relates to the trade-off between permissiveness and conservativeness; higher thresholds are likely too conservative and so too many states are left uncertain, perhaps missing true signals. Whereas lower thresholds may be too permissive and can lead to erroneous inferences. Interestingly, Keating (2023) highlights that uncertainty returned by a specified model is a better predictor of accuracy than likelihood diagnostics, such as AIC.

4.2.2 Sensitivity to fossil sampling

Given that uncertainty is often high in ASE, how then can estimates be improved? Ané (2008) demonstrates that ASE is *statistically inconsistent*, meaning that even if the sample size of extant tips is infinite, estimates are not expected to converge on the true value. Instead, the inclusion of fossils can improve estimates, due to the shorter evolutionary path to the root of the tree (Finarelli and Flynn, 2006; Finarelli and Goswami, 2013). Fossils can be used as priors on nodes, or simply added to the tree as tips.

By extension, inclusion of fossils can greatly influence inferences into disparity. Extant-only datasets often miss early rapid morphological diversification signals (Harmon et al., 2010), whereas analyses incorporating fossil data often finds the reverse (Hughes et al., 2013) For example, Mitchell (2015) found that analyses of bird disparity were underestimated when using ancestral state estimation of extant-only datasets.

4.2.3 Model dependency

Another key challenge with ASE is its sensitivity to violations of its model assumptions. For continuous characters, this is particularly pertinent when traits have evolved under non-BM (Revell, 2025). When trait evolution deviates from BM—due to directional trends, bounded evolution, or regime shifts—ASE can return biased estimations of ancestral states.

Both Webster and Purvis (2002) and Finarelli and Flynn (2006) were unable to recover accurate ancestral estimates of body size in planktonic foraminifera and mammals, respectively, due to a trend of increase through time. This violates the assumption of zero-mean change under BM. Similar issues arise when the trait data evolved under an Ornstein Uhlenbeck model, whereby trait data evolves in a Brownian way but under a stabilizing selection (Royer-Carenzi and Didier, 2016).

It is common for regime shifts to occur at mass extinction boundaries (Jablonski, 1986). For example, Slater (2013) found that mammal body size was relatively constrained (OU) up until the K-Pg mass extinction which then allowed a release of constraint and Brownian motion evolution.

As discussed previously, fossil sampling is crucial for accurate ASE, especially when model assumptions are violated (Puttick, 2016). Fossils assist in estimating and constraining model parameters, as well as distinguishing different evolutionary processes (Goswami and Clavel, 2025). For example, when estimating states under an OU model, fossils are necessary for estimating the primary optimum. Similarly, fossils would be necessary to detect a trend, such as Cope's rule. A similar logic applies to discrete characters.

Keating (2023) found that inclusion of fossils improved ASE estimates, irrespective of model misspecification.

4.2.4 Pre- or post-ordination

ASE can be performed at two points in the disparity analysis pipeline (Figure 3): either pre-ordination or post-ordination. Pre-ordination ASE involves estimating the ancestral states of the raw trait data, whereas post-ordination ASE estimates the PC scores of ancestral nodes. The pre-ordination has become favoured (Lloyd, 2018) due to various statistical issues with post-ordination ASE. For example, post-ordination ASE means ancestral values have to fall within the convex hull of the observed trait data (Stayton, 2015). While this is usually biologically realistic due to the zero-force evolutionary law (McShea and Brandon, 2010), in a scenario where taxon sampling is poor, true ancestral states may fall outside the morphological extremes designated by observed data. Lloyd (2018) found that the post-ordination ASE method recovers an implausibly low amount of convergent evolution and patterns dictated more so by phylogenetic signal than morphological

signal. Despite this, there are some practical benefits to performing post-ordination ASE. Firstly, it is less computationally expensive. Secondly, it mitigates issues with trait correlations. ASE methods assume trait independence, which is rare in morphological trait datasets (e.g., body size and limb length are positively correlated). Pre-ordination ASE, by ignoring this correlation, may return biologically implausible estimations that violate allometry, for example. Through ordination, the dominant sources of trait correlation are captured within the first few PC axes. The orthogonality of PC axes means zero co-variance between 'traits' (which are now PC axes) thus reducing the violation of the assumption of independence.

5 Detecting shifts in disparity

By splitting the morphospace into time slices, either by biostratigraphic slices or equidistant time slices by time slicing method. Therefore, can see how disparity changes over temporal scale. Often used in studies across mass extinction. How then to find if there is a significant shift in disparity through time? Well, MANOVA seems appealing to several studies. However (cite time series paper), simple averages do not tell full story - could just be a reflection of an upwards/downwards trend Therefore, need new techniques Smith (fungal paper) touched on this by simulating trait data as a way of showing the zero force evolutionary law (null model).

Part III

Thesis plan

6 Aims and objectives

Knowledge gap

The overarching aim of this PhD project is to understand how current methods in the disparity pipeline can be improved, with a focus on disparity analyses focusing on mass extinction events.

Research questions

- How robust are disparity analyses to uncertainty in ancestral state estimations?
- How do both selective and non-selective extinction events differ in the ability to recover accurate ancestral state estimations?
- Which of the three ecological hypotheses for the diversification of eutherian mammals is most strongly supported by disparity analyses?

Research objectives

- Develop a simulation procedure for assessing the accuracy of ancestral state estimation across a range of models of trait evolution, fossil sampling levels and disparity metrics.
- This simulation procedure can then be expanded to incorporate mass extinction events, with varying levels of intensity and selectivity.
- Construct several dated phylogenies for eutherian mammals, using both fossil and molecular data, each with different calibrations to essentially return long fuse, soft explosive and hard explosive trees.
- These trees can then be used in a disparity pipeline, using a dataset
-

7 Chapter One: Eutherian phylogenies

Chapter aim: To reconstruct the phylogeny of eutherian mammals, using various calibrations for the placental node in order to output 3 tree distributions: long fuse, soft explosive and hard explosive (subsection 2.4). The rationale of this chapter is so that these three tree distributions can be used in chapter four (section 10).
Research questions:

- When did major lineages of eutherians diversify?
- When did major shifts in evolutionary rates occur in eutherians?

Research objectives:

- I will use both molecular and morphological data for a total-evidence approach, which involves concatenating both molecular and morphological alignments into a single supermatrix
- The molecular and morphological matrices have been previously collected by my supervisor in a previous project
- Bayesian analysis will use RevBayes.
- I will implement different calibration strategies at the placental node to generate three alternative models: a long fuse model (placental node $\sim$ 100 Ma), a soft explosive scenario (placental node $\sim$ 80 Ma), and a hard explosive scenario (placental node 66 Ma).
- I will specify a Fossilized Birth-Death model in RevBayes to incorporate fossil tip ages and estimate divergence times.
- I will assess convergence and summarize the posterior distributions to obtain time-calibrated phylogenetic trees that reflect each calibration scenario.
- I will use RevGadgets to summarise the trees and plot figures.

8 Chapter Two: What is the effect of ancestral state estimations on disparity?

Chapter aim: To investigate how different approaches for treating uncertainties in ancestral state estimations affect estimates of morphological disparity under varying levels of fossil sampling. To expand this framework to understand how random and selective mass extinction events can confound ancestral state and disparity estimation accuracy.

Research questions:

- Which method of treating uncertainties recovers the most accurate ancestral state estimations?
- How does estimation accuracy change across different fossil sampling levels?
- How does estimation accuracy of disparity differ to accuracy of ancestral state estimation?
- What are the effects of mass extinction events on ancestral state estimations?
- How do random vs selective extinction events differ in their effect on ancestral state estimations?

Research objectives:

- Develop a simulation procedure that generates a distribution of 'true' evolutionary histories (i.e., phylogeny and trait data), with both discrete and continuous trait data.
- Remove varying levels of fossil tips from these 'true' phylogenies to simulate different fossil sampling levels.
- Run ancestral state estimation on these pruned trees/matrices with three different ways of treating uncertainty in estimates.
- Compare the raw distance between true vs estimated matrices.
- Compare the difference in disparity values between true vs estimated matrices.
- Run this procedure, but incorporating a mass extinction event into the simulations. These mass extinction events will have varying intensities and selectivities.

9 Chapter Three: Measuring changes in trends in disparity through time

Chapter aim: To develop a framework that can distinguish between random and non-random shifts in disparity through time, accounting for trend and stochasticity.

Research questions:

- How can traditional statistical tests, such as changes in average (MANOVA), be misleading in inferences of change in disparity?
- How to implement a BACI (Before After Control Intervention) approach into disparity through time?
- How to use simulated data as a valid Control?
- How to deal with time lags?

Research objectives:

- Using the BACI framework laid out by Wauchope et al. (2021) I will develop a method to quantify shifts in disparity across a time interval (say a mass extinction for example)
- This framework uses a before and
- Simulated data will be used as the null model (control). Simulated data will be fit using same parameters as the pre-intervention data.
- I will implement this on both simulated and empirical data
- I will

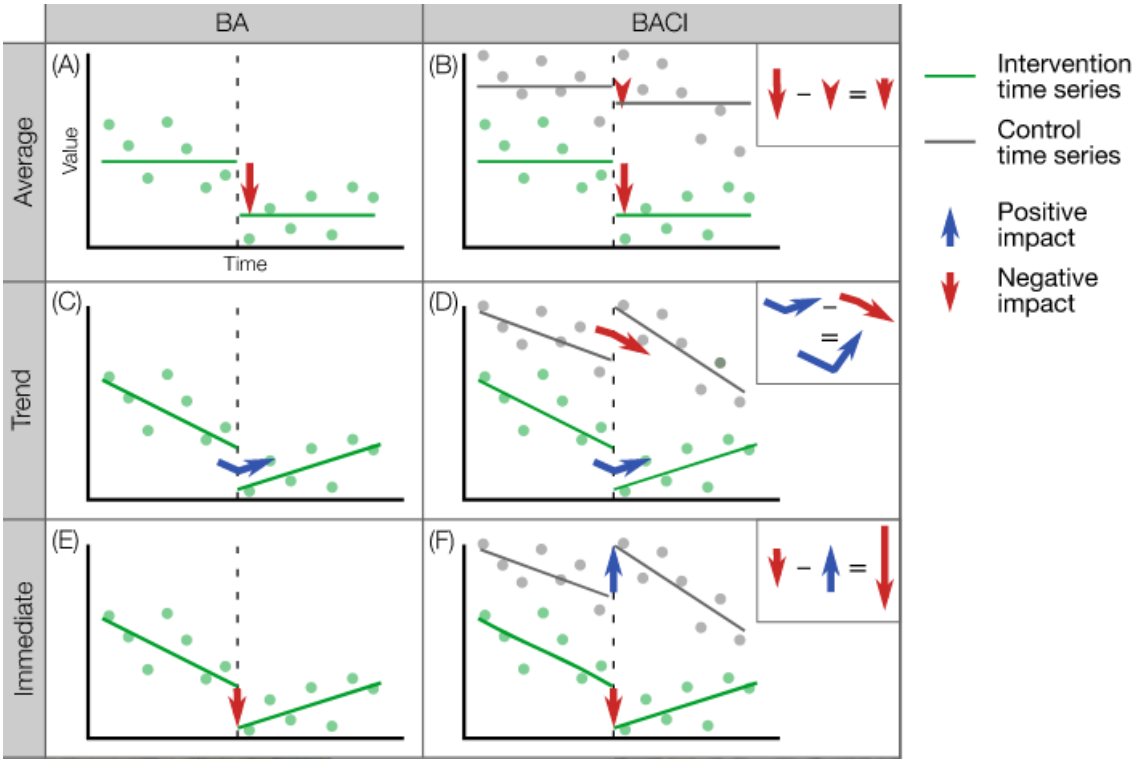


Figure 5: BA and BACI approach compared. Figure from Wauchope et al. (2021).

10 Chapter Four: Eutherian disparity over K-Pg mass extinction

Chapter aim: Incorporate the phylogenies from chapter one, the ancestral state method from chapter two and the test for non-random shifts in disparity from chapter two to develop a pipeline that can test whether the K-Pg mass extinction had an effect eutherian disparity.

Research questions:

- Is there a shift in trend of disparity at the K-Pg boundary?
- Does this shift depend on the phylogenetic topology used?
- Do patterns differ between different disparity metrics?

Research objectives:

- Use a dataset of discrete morphofunctional characters of jaw morphology and dentition, collected by Emmerson (2022).
- Develop a pipeline that iterates through different models and metrics: tree topology (long fuse/soft explosive/hard explosive), distance metric, time slicing model and disparity metric.
- From 100s of disparity curves, use the function developed in chapter three to score a value of 'change in disparity' at 66 Ma.
- Develop an interactive `shiny` app that presents the results in an 'electoral map' style format, whereby each 'pixel', representing a specific combination of topology, model and metric, is shaded on a scale of decrease to increase in disparity.

Part IV

Summary of research achievements

11 Introduction

- Morphology can tell us interesting patterns about evolutionary processes.
- The combination of morphological datasets (using fossils) and phylogenies means we can infer these processes through deep time
- This is particularly useful when studying major evolutionary events (such as a mass extinction)
- However, the fossil record is incomplete and this restricts our ability to look at deep time patterns of morphology.

- Many studies have made use of ancestral state estimation (ASE) to fill in gaps left by 'ghost lineages'. We know some species must have existed between root and tips but we don't know what they looked like.
- Particularly has been implemented in studies looking at disparity, which describes the multidimensional 'morphospace' in which taxa occupy.
- ASE can be used to estimate the character state values (both continuous and discrete characters) at nodes on a phylogeny, using observed trait data (from living and fossil specimens).

11.0.1 Why does this matter?

- ASE estimates are often shit and a 'glorified average'. But, what can we do with them?
- Erroneous estimates can mislead values of disparity and therefore macroevolutionary inferences.

11.1 Methods

11.1.1 Simulations

- I simulated a set of 100 trees using `treats` (Guillerme, 2024). These trees were simulated with birth-death parameters as $\lambda = 1.0$, and $\mu = 0.7$. This leaves trees with a range of 50-200 fossil tips and 50 extant tips.
- I simulated discrete and continuous trait data using `map.traits`. This effectively maps the 'true' evolutionary history across the phylogeny (tips and nodes have states mapped).
- For discrete traits, I simulated trait data across each tree (100 characters, 85% binary, 15% 3-state) using Mk model with 3 different transition rates: fast = 0.9, medium = 0.1, slow = 0.01.
- For continuous traits, I simulated BM and OU model trait data (50 characters).

11.1.2 Fossil sampling

- I then removed all node states from the matrices, leaving just tip states.
- In order to understand how ASE accuracy is affected by fossil sampling level, I pruned the fossil tips at 5 levels, preserving the root to maintain tree height across each level:
- **Living**: only extant tips are retained
- **Low**: 5% fossils retained
- **Medium**: 15% fossils retained
- **High**: 50% fossils retained

- **All**: All fossils retained

11.1.3 ASE

- I then ran ASE on these matrices using `multi.ace`, with three different parameters (for discrete states).
- **Relative majority rule**: the ancestral state estimated are all the ones that are have a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states:

$$\text{estimated state} \geq \left(\max(\text{likelihood}) - \frac{1}{n_{\text{states}}}\right)$$

  If more than one state is over this threshold, it is recovered as uncertain e.g., (0/1).
- **Strict majority rule**: Setting threshold parameter to a set value will select the ancestral states estimate with a scaled likelihood equal or greater than this designated value (in this case I did 0.5).
- **No majority rule (sampling)**: `multi.ace` has a `sample = x` parameter, whereby you can return x matrices, in which states returned are by the scaled likelihood of each state appearing at each node. In this case I used sample = 10 for fast computation (in future I will do 100/1000).
- Ran `multi.ace` on continuous characters using BM as model.
- I will also do post-ordination ace to see how that compares to pre-ordination ace.

11.1.4 Raw distance comparison between true vs ace

- I then used a Hamming distance metric to compare corresponding nodes between the 'true' vs 'aced' matrices.
- Distances were calculated only for nodes that were present in all fossil sampling levels to allow for comparison.
- The distance assigned a penalty of 1 for any wrong estimated state and also for uncertainty (i.e 0/1).
- For continuous traits, I firstly scaled each matrix and then measured the euclidean distance between true vs ace for each corresponding node.
- I also looked at the distance at just the root, to see how well ace recovers true value at deepest node.

11.1.5 Distance comparison between ordinated true vs ace

- I then ordinated the true matrix (distance matrix then PCO for discrete, PCA for continuous), then projected the aced matrices onto this true matrix.
- I measured the Euclidean distance between corresponding nodes again.

11.1.6 Difference in disparity between true vs ace

- from the ordinated matrices, I calculated (will choose which metrics, at the moment sum variance and ranges and average nearest neighbour distance) disparity and calculated the difference between true vs aced.

11.2 Results (or what I expect to find)

11.2.1 Raw distances

- Slower transition rates = ASE accuracy improves
- Increasing fossil sampling improves ASE accuracy (for discrete traits), particularly at deeper nodes.
- Using sample method of ASE improved estimate accuracy
- Using sample method is relatively insensitive to low fossil sampling and finds high accuracy at all fossil sampling levels
- At the root node, the impact of fossil sampling becomes more pertinent i.e., higher fossil sampling improves estimates
- For continuous traits, ASE accuracy was much better for BM traits than OU
- Estimates of continuous traits stay relatively constant across fossil preservation levels.
- For continuous traits, using point estimates was more accurate than sample method.

11.2.2 Ordination

- I expect that patterns of distance to true value will correspond to the raw distance pattern (more fossil sampling = higher ASE accuracy, sample method = higher ASE accuracy)
- I expect that fast transition rates will exhibit clustering at the centre of the morphospace, particularly with the default multi.ace parameter, due to the high amount of uncertainty recovered at deeper nodes. I think that the sample method of ASE will alleviate this artefact by removing this uncertainty.

11.2.3 Disparity

- Preliminary results have found that using sample ASE method recovers high accuracy for all transition rates.
- Using the default ASE method recovers much lower accuracy, particularly with low fossil sampling.

- I expect to see different results for different disparity metrics. For example, metrics that focus on density will be able to identify these artifacts of clustering due to uncertainty in node estimates, which variance based metrics will miss.
- This will highlight the need for multiple different disparity metrics to be used in disparity analyses

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